

Задачи по математическому анализу с РМО (включая оптимизацию)

1. (1995) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a surjective function with property that for any divergent sequence of real numbers $(x_n)_{n \geq 1}$, the sequence $(f(x_n))_{n \geq 1}$ is also divergent. Prove that f is continuous.

2. Let $M > 0$ be a real number and $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function with the property that

$$|f(x+y) - f(x) - f(y)| \leq M \quad \forall x, y \in \mathbb{R}.$$

Prove that for every $x \in \mathbb{R}$ the limit $\lim_{n \rightarrow \infty} \frac{f(nx)}{n}$ exists, denoted by $g(x)$, and the function $g : \mathbb{R} \rightarrow \mathbb{R}$ thus defined is continuous at 0. Deduce that the $\lim_{n \rightarrow \infty} \frac{f(x)}{x}$ exists and is finite.

3. Let $a < b$ be a real numbers and $f, g : [a, b] \rightarrow \mathbb{R}$ be two functions with the properties:

- a) The function f has antiderivatives and it is bounded above;
- b) For any $x, y \in [a, b]$, $x < y$ we have $g(y) - g(x) \geq (y - x) \sup_{t \in [x, y]} f(t)$.

Prove that g is integrable.

4. Let $M = \left\{ f : [0, \pi] \rightarrow \mathbb{R} \mid f \text{ is continuous and } \int_0^\pi f(x) \sin x dx = \int_0^\pi f(x) \cos x dx = 1 \right\}$. Find $\min_{f \in M} \int_0^\pi f^2(x) dx$.

5. (1996) Let $I \subset \mathbb{R}$ be a nondegenerate interval and $f : I \rightarrow \mathbb{R}$ a differentiable function.

We denote $J = \left\{ \frac{f(b)-f(a)}{b-a} : a, b \in I, a < b \right\}$. Prove that

- a) J is an interval;
- b) $J \subset f'(I)$, and the set $f'(I) \setminus J$ contains at most two elements;
- c) Using parts a) and b), deduce that f' has the intermediate value property.

6. a) Let $f_1, f_2, \dots, f_n : \mathbb{R} \rightarrow \mathbb{R}$ be periodic functions such that the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f = f_1 + f_2 + \dots + f_n$ has a finite limit at ∞ . Prove that f is constant.

b) If a_1, a_2, a_3 are real numbers such that $a_1 \cos(a_1 x) + a_2 \cos(a_2 x) + a_3 \cos(a_3 x) \geq 0$ for every $x \in \mathbb{R}$, then $a_1 a_2 a_3 = 0$.

7. Suppose that $f : [a, b] \rightarrow \mathbb{R}$ be a monotonic function and for every $x_1, x_2 \in [a, b]$ that $x_1 < x_2$, there exists $c \in (a, b)$ such that $\int_{x_1}^{x_2} f(x) dx = f(c)(x_2 - x_1)$.

- a) Show that f be the continuous function on interval (a, b)
- b) Suppose that f is integrable function on interval $[a, b]$ but f isn't a monotonic function then is it the result of part a) right?

8. Let $f : [0, 1) \rightarrow \mathbb{R}$ be a monotonic function. Prove that the limits

$$\lim_{x \nearrow 1} \int_0^x f(t) dt \text{ and } \lim_{n \rightarrow \infty} \frac{1}{n} \left(f(0) + f\left(\frac{1}{n}\right) + \cdots + f\left(\frac{n-1}{n}\right) \right)$$

exist and are equal.

9. (1997) Let $\mathcal{F}$ be the set of the differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f(x) \geq f(x + \sin x)$ for any $x \in \mathbb{R}$.

a) Prove that there exist nonconstant functions in $\mathcal{F}$.

b) Prove that if $f \in \mathcal{F}$, then the set of solutions of the equation $f'(x) = 0$ is infinite.

10. Let two bijective and continuous functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that: $(f \circ g^{-1})(x) + (g \circ f^{-1})(x) = 2x$ for any real x . Show that if we have a value $x_0 \in \mathbb{R}$ such that $f(x_0) = g(x_0)$, then $f = g$.

11. Prove that

$$\int_{-1}^1 f^2(x) dx \geq \frac{1}{2} \left(\int_{-1}^1 f(x) dx \right)^2 + \frac{3}{2} \left(\int_{-1}^1 x f(x) dx \right).$$

12. Suppose that $(f_n)_{n \in \mathbb{N}}$ be the sequence from all functions $f_n : [0, 1] \rightarrow \mathbb{R}^+$ s.t. f_0 be the continuous function and $\forall x \in [0, 1] \forall n \in \mathbb{N}, f_{n+1}(x) = \int_0^x \frac{1}{1 + f_n(t)} dt$. Prove that for every $x \in [0, 1]$ the sequence of $(f_n(x))_{n \in \mathbb{N}}$ be the convergent sequence and calculate the limitation.

13. (1998) Let $(a_n)_{n \geq 1}$ be a sequence of real numbers satisfying the properties:

13.1. the sequence $x_n = \sum_{k=1}^n a_k^2$ is convergent;

13.2. the sequence $y_n = \sum_{k=1}^n a_k$ is inbounded.

Prove that the sequence $(b_n)_{n \geq 1}$ given by $b_n = \{y_n\}$ is divergent.

14. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function for which the inequality $f'(x) \leq f'(x + \frac{1}{n})$ holds for every $x \in \mathbb{R}$ and every $n \in \mathbb{N}$. Prove that f is continously differentiable.

15. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function with the property that for any $a, b \in \mathbb{R}, a < b$, there are $c_1, c_2 \in [a, b]$, $c_1 \leq c_2$ such that $f(c_1) = \min_{x \in [a, b]} f(x)$ and $f(c_2) = \max_{x \in [a, b]} f(x)$. Prove that f is increasing.

16. Suppose that $a, b \in \mathbb{R}^+$ which $a + b < 1$ and $f : [0, +\infty) \rightarrow [0, +\infty)$ be the increasing function s.t

$$\forall x \geq 0, \int_0^x f(t) dt = \int_0^{ax} f(t) dt + \int_0^{bx} f(t) dt.$$

Prove that $\forall x \geq 0 f(x) = 0$.

17. (1999) Let $a, b \in \mathbb{R}, a < b$ and $f, g : [a, b] \rightarrow \mathbb{R}$ two differentiable functions with increasing derivatives and $f'(a) > 0, g'(a) > 0$. Prove that there exists $c \in [a, b]$ such that

$$\frac{f(b) - f(a)}{b - a} \cdot \frac{g(b) - g(a)}{b - a} = f'(c)g'(c).$$

18. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that

$$f(x) = f\left(\frac{x}{2}\right) + \frac{x}{2}f'(x) \quad \forall x \in \mathbb{R}.$$

Prove that f is polynomial function of degree at most one.

19. Find all continuous functions $f : \mathbb{R} \rightarrow [1, \infty)$ for which there exists $a \in \mathbb{R}$ and a positive integer k such that

$$f(x)f(2x) \cdots f(nx) \leq an^k$$

for all real x and all positive integers n .

20. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a monotonic function and a, b, c, d be real numbers with a and c nonzero. Prove that if the equalities

$$\int_x^{x+\sqrt{3}} f(t)dt = ax + b \text{ and } \int_x^{x+\sqrt{2}} f(t)dt = cx + d$$

hold for every real number x , then f is a polynomial function of degree one.

21. (2000) Study the convergence of sequence $(x_n)_{n \geq 0}$ for which $x_0 \in \mathbb{R} \setminus \mathbb{Q}$, and $x_{n+1} \in \left\{ \frac{x_n+1}{x_n}, \frac{x_n+2}{2x_n-1} \right\} \forall n \geq 1$.

22. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function that satisfies the conditions:

(i) $\lim_{x \rightarrow \infty} (f \circ f)(x) = \infty = - \lim_{x \rightarrow \infty} (f \circ f)(x);$

(ii) f has Darboux's property.

a) Prove that the limits of f at $\pm\infty$ exist.

b) Is possible for the limits from a) to be finite?

23. Let $a \in (1, \infty)$ and a continuous function $f : [0, \infty) \rightarrow \mathbb{R}$ having the property:

$$x \lim_{x \rightarrow \infty} f(x) \in \mathbb{R}.$$

a) Show that the integral $\int_1^\infty \frac{f(x)}{x} dx$ and the limit $\lim_{t \rightarrow \infty} t \int_1^a f(x^t) dx$ both exist, are finite and equal.

b) Calculate $\lim_{t \rightarrow \infty} t \int_1^a \frac{dx}{1+x^t}$.

24. For any partition P of $[0, 1]$, consider the set

$$\mathcal{A}(P) = \left\{ f : [0, 1] \rightarrow \mathbb{R} \mid \left| \exists f' \right|_{[0,1]} \wedge \int_0^1 |f(x)| dx = 1 \wedge (y \in P \implies f(y) = 0) \right\}.$$

Prove that there exists a partition P_0 of $[0, 1]$ such that

$$g \in \mathcal{A}(P_0) \implies \sup_{x \in [0,1]} |g'(x)| > 4 \cdot \#P.$$

Here $\#D$ denotes the natural number d such that $0 = x_0 < x_1 < \cdots < x_d = 1$ is a partition D of $[0, 1]$.

25. (2001) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ a continuous function, derivable on $\mathbb{R} \setminus x_0$, having finite side derivatives in x_0 . Show that there exists a derivable function $g : \mathbb{R} \rightarrow \mathbb{R}$, a linear function $h : \mathbb{R} \rightarrow \mathbb{R}$ and $\alpha \in \{-1, 0, 1\}$ such that

$$f(x) = g(x) + \alpha|h(x)| \quad \forall x \in \mathbb{R}.$$

26. Let $f : \mathbb{R} \rightarrow [0, +\infty)$ be a function with the property that $|f(x) - f(y)| \geq |x - y|$ for every $x, y \in \mathbb{R}$. Show that

- a) If $\lim_{n \rightarrow \infty} f(x + n) = \infty$ for every $x \in \mathbb{R}$, then $\lim_{x \rightarrow \infty} f(x) = \infty$.
b) If $\lim_{n \rightarrow \infty} f(x + n) = \alpha$ for every $x \in \mathbb{R}$, then $\lim_{x \rightarrow \infty} f(x) = \alpha$.

27. The continuous function $f : [0, 1] \rightarrow \mathbb{R}$ has the property:

$$\lim_{n \rightarrow \infty} n \left(n f \left(x + \frac{1}{n} \right) - f(x) \right) = 0$$

for every $x \in [0, 1)$. Show that

- a) For every $\varepsilon > 0$ and $\lambda \in (0, 1)$ we have

$$\sup_{x \in [0, \lambda)} |f(x) - f(0)| \leq \varepsilon x = \lambda.$$

- b) f is a constant function.

28. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a continuous function. Show that:

- a) if $\int_0^1 f(\sin(x + \alpha)) dx = 0$ for every $\alpha \in \mathbb{R}$, then $f(x) = 0 \quad \forall x \in [-1, 1]$.
b) if $\int_0^1 f(\sin(nx)) dx = 0$ for every $n \in \mathbb{Z}$, then $f(x) = 0 \quad \forall x \in [-1, 1]$.

29. Let $f : [0, +\infty) \rightarrow \mathbb{R}$ be a periodical function with period 1, integrable on $[0, 1]$. For a strictly increasing and unbounded sequence $(x_n)_{n \geq 0}$, $x_0 = 0$, with $\lim(x_{n+1} - x_n) = 0$, we denote $r(n) = \max\{k | x_k \leq n\}$.

- a) Show that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^{r(n)} (x_k - x_{k+1}) f(x_k) = \int_0^1 f(x) dx.$$

- b) Show that

$$\lim_{n \rightarrow \infty} \frac{1}{\ln n} \sum_{k=1}^{r(n)} \frac{f(\ln k)}{k} = \int_0^1 f(x) dx.$$

30. (2002) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function that has limits at any point and has no local extrema. Show that

- a) f is continuous;
b) f is strictly monotone.

31. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous and bijective function. Describe the set:

$$A = \{f(x) - f(y) | x, y \in [0, 1] \setminus \mathbb{Q}\}$$

32. Let $f : [0, 1] \rightarrow \mathbb{R}$ be an integrable function such that:

$$0 < \left| \int_0^1 f(x) dx \right| \leq 1.$$

Show that there exists $x_1 \neq x_2$, $x_1, x_2 \in [0, 1]$ such that

$$\int_{x_1}^{x_2} f(x) dx = (x_1 - x_2)^{2002}.$$

33. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous and bounded function such that

$$x \int_x^{x+1} f(t) dt = \int_0^\pi f(t) dt \text{ for any } x \in \mathbb{R}.$$

Prove that f is a constant function.

34. (2003) Let be two functions $f, g : \mathbb{R}_+ \rightarrow \mathbb{R}$ having that properties that f is continuous, g is nondecreasing and ingounded, and for any sequence of rational numbers $(x_n)_{n \geq 1}$ that diverges to ∞ , we have

$$1 = \lim_{n \rightarrow \infty} f(x_n)g(x_n).$$

Prove that $1 = \lim_{x \rightarrow \infty} f(x)g(x)$.

35. Let be an odd natural number $n \geq 3$. Find all continuous functions $f : [0, 1] \rightarrow \mathbb{R}$ that satisfy the following equalities:

$$\int_0^1 (f(\sqrt[k]{x}))^{n-k} dx = \frac{k}{n} \quad \forall k \in \{1, 2, \dots, n-1\}.$$

36. Let be a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ that has the property that

$$xf(x) \geq \int_0^x f(t) dt$$

for all real numbers x . Prove that

a) the mapping $x \rightarrow \frac{1}{x} \int_0^x f(t) dt$ is nondecreasing on the restrictions $\mathbb{R}_{<0}$ and $\mathbb{R}_{>0}$.

b) if $\int_x^{x+1} f(t) dt = \int_{x-1}^x f(t) dt$ for any real number x , then f is constant.

37. (2004) Let $f : (a, b) \rightarrow \mathbb{R}$ be a function with property that for all $x \in (a, b)$ there is a non-degenerated interval $[a_x, b_x]$ with $a < a_x \leq x \leq b_x < b$ such that f is constant on $[a_x, b_x]$.

a) Prove that $\text{Im } f$ is finite or numerable.

b) Find all continuous functions which have the property mentioned in the hypothesis.

38. a) Build a function $f : \mathbb{R} \rightarrow \mathbb{R}_+$ with property $(\mathcal{P})$, i.e, all $x \in \mathbb{Q}$ are local, strict minimum points

b) Build a function $f : \mathbb{Q} \rightarrow \mathbb{R}_+$ such that every point is a local, strict minimym point and such that f is unbounded on $I \cap \mathbb{Q}$, where I is a non-degenerate interval.

- c) Let $f : \mathbb{R} \rightarrow \mathbb{R}_+$ be a function unbounded on every $I \cap \mathbb{Q}$, where I is non-degenerate interval. Prove that f doesn't have the property $(\mathcal{P})$.

39. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all $x \in \mathbb{R}$ and for all $n \in \mathbb{N}^*$ we have

$$n^2 \int_x^{x+\frac{1}{n}} f(t) dt = nf(x) + \frac{1}{2}.$$

40. Let $f : [0, 1] \rightarrow \mathbb{R}$ be an integrable function such that

$$\int_0^1 f(x) dx = \int_0^1 xf(x) dx = 1.$$

Prove that

$$\int_0^1 f^2(x) dx \geq 4.$$

41. (2005) Let $f : [0, 1] \rightarrow (0, 1)$ a continuous onto (surjective) function.

- Prove that for all $a \in (0, 1)$ the function $f_a : (a, 1) \rightarrow (0, 1)$ given by $f_a(x) = f(x)$ for all $x \in (a, 1)$ is onto;
- Give an example of such a function.

42. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function.

- Prove that f is continuous;
- Prove that there exists a unique function $g : [0, \infty) \rightarrow \mathbb{R}$ such that for all $x \geq 0$ we have

$$f(x + g(x)) = f(g(x)) - g(x).$$

43. Let $f : [0, \infty) \rightarrow (0, \infty)$ a continuous function such that $\lim_{n \rightarrow \infty} \int_0^n f(t) dt$ exists and it is finite. Prove that

$$\lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} \int_0^x \sqrt{f(t)} dt = 0.$$

44. (2006) Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a function such that for any $x > 0$ the sequence $\{f(nx)\}_{n \geq 0}$ is increasing.

- If the function is also continuous on $[0, 1]$ is it true that f is increasing?
- The same question if the function is continuous on $\mathbb{R} \cup [0, \infty)$

45. Prove that

$$\lim_{n \rightarrow \infty} n \left(\frac{\pi}{4} - n \int_0^1 \frac{x^n}{1 + x^{2n}} dx \right) = \int_0^1 f(x) dx,$$

where $f(x) = \frac{\arctan x}{x}$ if $x \in (0, 1]$ and $f(0) = 1$.

46. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that

$$\int_0^1 f(x) dx = 0.$$

Prove that there is $c \in (0, 1)$ such that

$$\int_0^c xf(x)dx = 0.$$

47. (2007) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function, and $a < b$ be two points in the image of f . Show that there is an interval such that $f(I) = [a, b]$.

48. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative, that satisfies $f(x + f'(x)) = f(x)$. Let's call this property (P) .

- a) Show that if f is a function with property (P) , then there exists a real x such that $f'(x) = 0$;
- b) Give an example of a non-constant function f with property (P) ;
- c) Show that if f has property (P) and the equation $f'(x) = 0$ has at least two solutions, then f is a constant function.

49. Let $\mathcal{F}$ be the set of functions $f : [0, 1] \rightarrow \mathbb{R}$ that are differentiable, with continuous derivative, and $f(0) = 0$, $f(1) = 1$. Find the minimum of

$$\int_0^1 \sqrt{1+x^2} \cdot (f'(x))^2 dx$$

(where $f \in \mathcal{F}$) and find all functions $f \in \mathcal{F}$ for which this minimum is attained.

50. Let $f : [0, 1] \rightarrow (0, +\infty)$ be a continuous function

- a) Show that for any integer $n \geq 1$, there is a unique division $0 = a_0 < a_1 < \dots < a_n = 1$ such that

$$\int_{a_k}^{a_{k+1}} f(x)dx = \frac{1}{n} \int_0^1 f(x)dx$$

holds for all $k = 0, 1, \dots, n-1$.

- b) For each n consider the a_i above (that depend on n and define $b_n = \frac{a_1 + a_2 + \dots + a_n}{n}$). Show that the sequence (b_n) is convergent and compute its limit.

51. (2008) Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that the sequences $\{f(nx)\}_{n \geq 1}$ are nondecreasing for any real number x . Prove that f is nondecreasing.

52. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function, two times derivable on $\mathbb{R}$ for which there exist $c \in \mathbb{R}$ such that

$$\frac{f(b) - f(a)}{b - a} \neq f'(c),$$

for all $a \neq b \in \mathbb{R}$. Prove that $f''(c) = 0$.

53. Let $a > 0$ and $f : [0, \infty) \rightarrow [0, a]$ be a continuous function on $(0, \infty)$ and having Darboux property on $[0, \infty)$. Prove that if $f(0) = 0$ and for all nonnegative x we have

$$xf(x) \geq \int_0^x f(t)dt,$$

then f admits primitives on $[0, \infty)$.

54. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a derivable function, with a continuous derivative f' on $[0, 1]$. Prove that if $f(\frac{1}{2}) = 0$, then

$$\int_0^1 (f'(x))^2 dx \geq 12 \left(\int_0^1 f(x) dx \right)^2.$$

55. (2009) Let $(t_n)_n$ a convergent sequence of real numbers, $t_m \in (0, 1) \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} t_n \in (0, 1)$. Define the sequence $(x_n)_n$ and $(y_n)_n$ by

$$x_{n+1} = t_n x_n + (1 - t_n) y_n,$$

$$y_{n+1} = (1 - t_n) x_n + t_n y_n, \quad \forall n \in \mathbb{N}$$

and x_0, y_0 are given real numbers.

a) Prove that the sequences $(x_n)_n$ and $(y_n)_n$ are convergent and have the same limit

b) Prove that if $\lim_{n \rightarrow \infty} t_n \in \{0, 1\}$, then the question is false.

56. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ a continuous function such that for any $x \in \mathbb{R}$ the limit $\lim_{h \rightarrow 0} \left| \frac{f(x+h)-f(x)}{h} \right|$ exists and is finite. Prove that in any real point, f is differentiable or it has finite one-side derivates, of the same modul, but different signs.

57. Let $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$ such that f is differentiable, g and h are monotonic, and $f' = f + g + h$. Prove that the set of the points of discontinuity of g coincides with the respective set of h .

58. Find all functions $f \in C^1[0, 1]$ that satisfy $f(1) = -\frac{1}{6}$ and

$$\int_0^1 (f'(x))^2 dx \leq 2 \int_0^1 f(x) dx.$$

59. Find all functions $f : [0, 1] \rightarrow [0, 1]$ that are bijective, continuous and have the property that for any continuous function $g : [0, 1] \rightarrow \mathbb{R}$ the following equality holds:

$$\int_0^1 g(f(x)) dx = \int_0^1 g(x) dx.$$

60. (2010) Let $f : \mathbb{R} \rightarrow [0, \infty)$. Prove that $f(x + y) \geq (y + 1)f(x) \ (\forall)x \in \mathbb{R} \iff$ the function $g(x) : \mathbb{R} \rightarrow [0, \infty)$, $g(x) = e^{-x}f(x)$, $(\forall)x \in \mathbb{R}$ is increasing.

61. Let $a \in \mathbb{R}_+$ and define the sequence of real numbers $(x_n)_n$ by $x_1 = a$ and $x_{n+1} = |x_n - \frac{1}{n}|$, $n \geq 1$. Prove that the sequence is convergent and find it's limit.

62. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a monotonic function and $F : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$F(x) = \int_0^x f(t) dt.$$

Prove that if F has a finite derivative, then f is continuous.

63. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a continuous function having finite derivative at 0, and

$$I(h) = \int_{-h}^h f(x) dx, \quad h \in [0, 1].$$

Prove that

- a) there exists $M > 0$ such that $|I(h) - 2f(0)h| \leq Mh^2$ for any $h \in [0, 1]$;
 - b) the sequence $(a_n)_{n \geq 1}$, defined by $a_n = \sum_{k=1}^n \sqrt{k} |I(1/k)|$ is convergent if and only if $f(0) = 0$.
64. (2011) Let $u : [a, b] \rightarrow \mathbb{R}$ be a continuous function that has finite left-side derivative $u'_l(x)$ in any point $x \in (a, b]$. Prove that the function u is monotonously increasing if and only if $u'_l(x) \geq 0$ for any $x \in (a, b]$.
65. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous and strictly decreasing function with $g(\mathbb{R}) = (-\infty, 0)$. Prove that there are no continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ with the property that there exists a natural number $k \geq 2$ so that:

$$f \circ f \circ \dots \circ f (k \text{ times}) = g.$$

66. Let be a continuous function $f : [0, 1] \rightarrow (0, \infty)$ having the property that for any natural number $n \geq 2$, there exist $n - 1$ real numbers $0 < t_1 < t_2 < \dots < t_{n-1} < 1$, such that

$$\int_0^{t_1} f(t)dt = \int_{t_1}^{t_2} f(t)dt = \int_{t_2}^{t_3} f(t)dt = \dots = \int_{t_{n-2}}^{t_{n-1}} f(t)dt = \int_{t_{n-1}}^1 f(t)dt.$$

Calculate $\lim_{n \rightarrow \infty} \frac{n}{\frac{1}{f(0)} + \sum_{i=1}^{n-1} \frac{1}{f(t_i)} + \frac{1}{f(1)}}.$

67. Let $f, F : \mathbb{R} \rightarrow \mathbb{R}$ be two functions such that f is nondecreasing, F admits finite lateral derivates in every point of its domain,

$$\lim_{x \rightarrow y-} f(x) \leq \lim_{x \rightarrow y-} \frac{F(x) - F(y)}{x - y}, \quad \lim_{x \rightarrow y+} f(x) \geq \lim_{x \rightarrow y+} \frac{F(x) - F(y)}{x - y}$$

for real numbers y , and $F(0) = 0$. Prove that $F(x) = \int_0^x f(t)dt$ for all real numbers x .

68. (2012) Let $f, g : [0, 1] \rightarrow [0, 1]$ be two functions such that g is monotonic, surjective and $|f(x) - f(y)| \geq |g(x) - g(y)|$ for any $x, y \in [0, 1]$.
- a) Prove that f is continuous and that there exists some $x_0 \in [0, 1]$ with $f(x_0) = g(x_0)$;
 - b) Prove that the set $\{x \in [0, 1] | f(x) = g(x)\}$ is a closed interval.

69. Find all differentiable functions $f : [0, \infty) \rightarrow [0, \infty)$ for which $f(0) = 0$ and $f'(x^2) = f(x)$ for any $x \in [0, \infty)$.

70. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that $\int_0^n f(x)f(n-x)dx = \int_0^n f^2(x)dx$ for any natural number $n \geq 1$. Prove that f is periodic function.

71. Let $\mathcal{C}$ be the set of integrable functions $f : [0, 1] \rightarrow \mathbb{R}$ such that $0 \leq f(x) \leq x$ for any $x \in [0, 1]$. Define the function $V : \mathcal{C} \rightarrow \mathbb{R}$ by

$$V(f) = \int_0^1 f^2(x)dx - \left(\int_0^2 f(x)dx \right)^2, \quad f \in \mathcal{C}$$

Determine the following two sets:

- a) $\{V(f_a) | 0 \leq a \leq 1\}$, where $f_a(x) = 0$ if $0 \leq x \leq a$ and $f_a(x) = x$ if $a < x \leq 1$;
- b) $\{V(f) | f \in \mathcal{C}\}.$

72. (2013) A function

$$f : (0, \infty) \rightarrow (0, \infty)$$

is called contract if, for every numbers $x, y \in (0, \infty)$ we have $\lim_{n \rightarrow \infty} (f^n(x) - f^n(y)) = 0$, where $f^n = \underbrace{f \circ f \circ \dots \circ f}_n$.

a) Consider

$$f : (0, \infty) \rightarrow (0, \infty)$$

a contract, continuous function with the property that it has a fixed point, i.e. there exists $x_0 \in (0, \infty)$ such that $f(x_0) = x_0$. Show that $f(x) > x$, for every $x \in (0, x_0)$ and $f(x) < x$, for every $x \in (x_0, \infty)$.

b) Show that the function

$$f : (0, \infty) \rightarrow (0, \infty)$$

given by $f(x) = x + \frac{1}{x}$ is contract but has no fixed point.

73. a) Consider

$$f : [0, \infty) \rightarrow [0, \infty)$$

a differentiable and convex function. Show that if $f(x) \leq x$, for every $x \geq 0$, then $f'(x) \leq 1$, for every $x \geq 0$.

b) Determine all differentiable and convex functions

$$f : [0, \infty) \rightarrow [0, \infty)$$

such that $f(0) = 0$, and $f'(x) \cdot f(f(x)) = x$, for every $x \geq 0$.

74. Determine continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $(a^2 + ab + b^2) \int_a^b f(x) dx = 3 \int_a^b x^2 f(x) ds$ for every $a, b \in \mathbb{R}$.

75. Given $a \in (0, 1)$ and C the set of increasing functions $f : [0, 1] \rightarrow [0, \infty)$ such that $\int_0^1 f(x) dx = 1$. Determine:

a) $\max_{f \in C} \int_0^a f(x) dx$

b) $\max_{f \in C} \int_0^a f^2(x) dx$

76. (2014) Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy:

(i) $\text{id} + f$ is nondecreasing,

(ii) there is a natural number m such that $\text{id} + f + f^2 + \dots + f^m$ is nonincreasing.

Here, id represents the identity function, and the exponent denotes functional power.

77. Find all derivable functions that have real domain and codomain, and are equal to their second functional power.

78. Let I, J be two intervals, $\varphi : J \rightarrow \mathbb{R}$ be a continuous function whose image doesn't contain 0, and $f, g : I \rightarrow J$ be two differentiable functions such that $f' = \varphi \circ f$, $g' = \varphi \circ g$ and such that the image of $f - g$ contains 0. Show that f and g are the same function.

79. Let $f : [1, \infty) \rightarrow (0, \infty)$ be a continuous function satisfying the following properties:

$$(i) \exists \lim_{x \rightarrow \infty} \frac{f(x)}{x} \in \mathbb{R}$$

$$(ii) \exists \lim_{x \rightarrow \infty} \frac{1}{x} \int_1^x f(t) dt \in \mathbb{R}.$$

$$a) \text{ Show that } \lim_{x \rightarrow \infty} \frac{f(x)}{x} = 0.$$

$$b) \text{ Prove that } \lim_{x \rightarrow \infty} \frac{1}{x^2} \int_1^x f^2(t) dt = 0.$$

80. (2015) Find all differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that verify the conditions:

$$(i) \forall a \in \mathbb{Z}, f'(a) = 0,$$

$$(ii) \forall x \in \mathbb{R}, f(x) = 0 \Rightarrow f'(x) = 0.$$

81. Let a, b be two nonnegative real numbers with $b > a$, and let $(x_n)_{n \geq 1}$ be a sequence of real numbers such that the sequence

$$\left(\frac{x_1 + x_2 + \cdots + x_n}{n^a} \right)_{n \geq 1}$$

is bounded.

Show that the sequence

$$\left(x_1 + \frac{x_2}{2^b} + \frac{x_3}{3^b} + \cdots + \frac{x_n}{n^b} \right)_{n \geq 1}$$

is convergent.

82. Let C be the set of all twice differentiable functions $f : [0, 1] \rightarrow \mathbb{R}$ with at least two (not necessarily distinct) zeros and $|f''(x)| \leq 1$, for all $x \in [0, 1]$. Find the greatest value of the integral

$$\int_0^1 |f(x)| dx$$

when f runs through the set C , as well as the functions that achieve this maximum.

Note: A differentiable function f has two zeros at the same point a if $f(a) = f'(a) = 0$.

83. Find all non-constant polynomials $f \in \mathbb{Q}[X]$ that do not have any real roots in the interval $[0, 1]$ and for which there exists a function

$$\varepsilon : [0, 1] \rightarrow \mathbb{Q}[X] \times \mathbb{Q}[X], \quad \varepsilon(x) := (g_x, h_x)$$

such that $h_x(x) \neq 0$ and

$$\int_0^x f(t) dt = \frac{g_x(x)}{h_x(x)} \quad \text{for all } x \in [0, 1].$$

84. (2016) Let a be a real number, and $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ a function. Show that the following relations are equivalent:

(i) For every $\varepsilon > 0$,

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x^{a+\varepsilon}} = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} \frac{f(x)}{x^{a-\varepsilon}} = \infty.$$

(ii)

$$\lim_{x \rightarrow \infty} \frac{\ln f(x)}{\ln x} = a.$$

85. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that f^2 is differentiable and $f = (f^2)'$.

86. Prove that there exists a unique sequence $(c_n)_{n \geq 1}$ of real numbers from the interval $(0, 1)$ such that

$$\int_0^1 \frac{dx}{1+x^m} = \frac{1}{1+c_m^m}, \quad \text{for all natural numbers } m,$$

and calculate $\lim_{k \rightarrow \infty} k c_k^k$.

87. Let a be a real number, and $f : \mathbb{R} \rightarrow \mathbb{R}$ a nondecreasing function. Prove that f is continuous at a if and only if there exists a sequence $(a_n)_{n \geq 1}$ of positive real numbers such that

$$\int_a^{a+a_n} f(x) dx + \int_a^{a-a_n} f(x) dx \leq \frac{a_n}{n}, \quad \text{for all natural numbers } n.$$

88. (2017) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a surjective function with the property that if the sequence $(f(x_n))_{n \geq 1}$ is convergent, then the sequence $(x_n)_{n \geq 1}$ is convergent. Prove that f is continuous.

89. Let f be a function of class $C^1[a, b]$ with positive derivative. Prove that there exists a real number $c \in (a, b)$ such that

$$f(f(b)) - f(f(a)) = f'(c)(f(b) - f(a)).$$

90. a) Let $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{> 0}$ be a continuous function. Show that there exists a natural number n_0 and a sequence of positive real numbers $(x_n)_{n > n_0}$ satisfying

$$n \int_0^{x_n} f(t) dt = 1, \quad \text{for all } n > n_0.$$

b) Prove that the sequence $(nx_n)_{n > n_0}$ is convergent and find its limit.

91. A function $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ satisfies

$$\lim_{x \rightarrow \infty} \frac{1}{x^2} \int_0^x f(t) dt = 1.$$

a) Give an example of such an f that is continuous and for which $\frac{f(x)}{x}$ does not have a limit at ∞ .

b) Prove that if f is nondecreasing, then $\frac{f(x)}{x}$ has a limit at ∞ , and determine this limit.

92. (2018) Let $x > 0$. Prove that

$$2^{-x} + 2^{-1/x} \leq 1.$$

93. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function with the intermediate value property. If f is injective on $\mathbb{R} \setminus \mathbb{Q}$, prove that f is continuous on $\mathbb{R}$.

94. Let $\mathcal{F}$ be the set of continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$e^{f(x)} + f(x) \geq x + 1, \quad \forall x \in \mathbb{R}.$$

For $f \in \mathcal{F}$, let

$$I(f) = \int_0^e f(x) dx.$$

Determine $\min_{f \in \mathcal{F}} I(f)$.

95. Let $f : [a, b] \rightarrow \mathbb{R}$ be an integrable function and $(a_n) \subset \mathbb{R}$ such that $a_n \rightarrow 0$.

- a) If $A = \{m \cdot a_n \mid m, n \in \mathbb{N}^*\}$, prove that every open interval of strictly positive real numbers contains elements from A .
- b) If, for any $n \in \mathbb{N}^*$ and for any $x, y \in [a, b]$ with $|x - y| = a_n$, the inequality

$$\left| \int_x^y f(t) dt \right| \leq |x - y|$$

holds, prove that

$$\left| \int_x^y f(t) dt \right| \leq |x - y|, \quad \forall x, y \in [a, b].$$

96. (2019) Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous function, constant on $\mathbb{Z}_{\geq 0}$. For any $0 \leq a < b < c < d$ satisfying $f(a) = f(c)$ and $f(b) = f(d)$, we also have

$$f\left(\frac{a+b}{2}\right) = f\left(\frac{c+d}{2}\right).$$

Prove that f is constant.

97. a) Prove that there exists a differentiable function $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(f'(x)) = x, \quad \forall x > 0.$$

b) Prove that there is no differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f'(x)) = x, \quad \forall x \in \mathbb{R}.$$

98. Let $a > 0$ and

$$\mathcal{F} = \{f : [0, 1] \rightarrow \mathbb{R} \mid f \text{ is concave and } f(0) = 1\}.$$

Determine

$$\min_{f \in \mathcal{F}} \left(\int_0^1 f(x) dx - (a+1) \int_0^1 x f(x) dx \right).$$

99. Let $f : [0, \infty) \rightarrow (0, \infty)$ be an increasing function and $g : [0, \infty) \rightarrow \mathbb{R}$ be a twice differentiable function such that g'' is continuous and

$$g''(x) + f(x)g(x) = 0, \quad \forall x > 0.$$

- a) Provide an example of such functions, with $g \not\equiv 0$.
- b) Prove that g is bounded.

100. (2021) Let $f : [a, b] \rightarrow \mathbb{R}$ be a function with the intermediate value property such that $f(a) \cdot f(b) < 0$. Show that there exist α, β with $a < \alpha < \beta < b$ such that

$$f(\alpha) + f(\beta) = f(\alpha) \cdot f(\beta).$$

101. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function of class C^n with $n \geq 2$ such that

$$\lim_{x \rightarrow \infty} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow \infty} f^{(n)}(x) = 0.$$

Prove that

$$\lim_{x \rightarrow \infty} f^{(k)}(x) = 0 \quad \text{for all } k \in \{1, 2, \dots, n-1\},$$

where $f^{(k)}$ denotes the k -th derivative of f .

102. Find all continuous functions $f : [0, 1] \rightarrow [0, \infty)$ such that

$$\int_0^1 f(x) dx \cdot \int_0^1 f^2(x) dx \cdots \int_0^1 f^{2020}(x) dx = \left(\int_0^1 f^{2021}(x) dx \right)^{1010}.$$

103. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous and bijective function such that $f(0) = 0$. Prove that the following inequality holds:

$$(\alpha + 2) \cdot \int_0^1 x^\alpha (f(x) + f^{-1}(x)) dx \leq 2, \quad \forall \alpha \geq 0.$$

104. (2022) Let $f : [0, 1] \rightarrow (0, 1)$ be a surjective function.

- Prove that f has at least one point of discontinuity.
- Given that f admits a limit at any point of the interval $[0, 1]$, show that it has at least two points of discontinuity.

105. Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which are differentiable at 0 and satisfy the following inequality for all real numbers x, y :

$$f(x + y) + f(xy) \leq f(x) + f(y).$$

106. Let $\mathcal{F}$ be the set of functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(2x) = f(x)$ for all $x \in \mathbb{R}$.

- Determine all functions $f \in \mathcal{F}$ which admit antiderivatives on $\mathbb{R}$.
- Give an example of a non-constant function $f \in \mathcal{F}$ which is integrable on any interval $[a, b] \subset \mathbb{R}$ and satisfies

$$\int_a^b f(x) dx = 0$$

for all real numbers a and b .

107. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two nondecreasing functions.

a. Show that for any $a \in \mathbb{R}$, $b \in [f(a-0), f(a+0)]$ and $x \in \mathbb{R}$, the following inequality holds:

$$\int_a^x f(t) dt \geq b(x-a).$$

b. Given that $[f(a-0), f(a+0)] \cap [g(a-0), g(a+0)] \neq \emptyset$ for any $a \in \mathbb{R}$, prove that for any real numbers $a < b$:

$$\int_a^b f(t) dt = \int_a^b g(t) dt.$$

Note: $h(a-0)$ and $h(a+0)$ denote the left-hand and right-hand limits, respectively, of a function h at point $a \in \mathbb{R}$.

108. (2023) Determine all twice differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$(f'(x))^2 + f(x) < 0, \quad \forall x \in \mathbb{R}.$$

109. Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$ for which there exist a differentiable function $g : \mathbb{R} \rightarrow \mathbb{R}$ and a sequence $(a_n)_{n \geq 1}$ of positive real numbers converging to 0 such that

$$g'(x) = \lim_{n \rightarrow \infty} \frac{f(x+a_n) - f(x)}{a_n}, \quad \forall x \in \mathbb{R}.$$

a) Give an example of such a function f that is not differentiable at any point $x \in \mathbb{R}$.

b) Show that if f is continuous on $\mathbb{R}$, then f is differentiable on $\mathbb{R}$.

110. Let $a, b \in \mathbb{R}$ with $a < b$. We say that $f : [a, b] \rightarrow \mathbb{R}$ has property (P) if there exists an integrable function on $[a, b]$ such that

$$f(x) - f\left(\frac{x+a}{2}\right) = f\left(\frac{x+b}{2}\right) - f(x), \quad \forall x \in [a, b].$$

Show that for every real number t there exists a unique function $f : [a, b] \rightarrow \mathbb{R}$ with property (P) satisfying

$$\int_a^b f(x) dx = t.$$

111. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a non-decreasing function of class C^1 with $f(0) = 0$. Define $g : [0, 1] \rightarrow \mathbb{R}$ by

$$g(x) = f(x) + (x-1)f'(x), \quad \forall x \in [0, 1].$$

a) Show that

$$\int_0^1 g(x) dx = 0.$$

b) Prove that for every convex differentiable function $\phi : [0, 1] \rightarrow [0, 1]$ with $\phi(0) = 0$ and $\phi(1) = 1$, the following inequality holds:

$$\int_0^1 g(\phi(t)) dt \leq 0.$$

112. (2024) Let $I \subset \mathbb{R}$ be an open interval and $f : I \rightarrow \mathbb{R}$ a twice differentiable function such that $f(x)f''(x) = 0$ for any $x \in I$. Prove that $f''(a) = 0$ for any $a \in I$.

113. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be functions with $g(x) = 2f(x) - f(2x)$ for all $x \in \mathbb{R}$.

- a) Prove that, if f is bounded in a neighbourhood of the origin and g is continuous at the origin, then f is continuous at the origin.
- b) Provide an example of a function f , discontinuous at the origin, for which the function g is continuous at the origin.

114. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $f(x) + \sin(f(x)) \geq x$ for all $x \in \mathbb{R}$. Prove that

$$\int_0^\pi f(x) dx \geq \frac{\pi^2}{2} - 2.$$

115. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function with $f(1) = 0$. Prove that the limit

$$\lim_{t \rightarrow 1} \left(\frac{1}{1-t} \int_0^1 x(f(tx) - f(x)) dx \right)$$

exists and find its value.

116. (2025) Find all pairs of twice differentiable functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$, with their second derivatives being continuous, such that for all $x, y \in \mathbb{R}$:

$$(f(x) - g(y))(f'(x) - g'(y))(f''(x) - g''(y)) = 0.$$

117. Prove that for a function $f : \mathbb{R} \rightarrow \mathbb{R}$, the following two statements are equivalent:

- a) f is differentiable with continuous first derivative.
- b) For any $a \in \mathbb{R}$ and any two sequences $(x_n)_{n \geq 1}, (y_n)_{n \geq 1}$ convergent to a , with $x_n \neq y_n$ for all $n \in \mathbb{N}^*$, the sequence

$$\left(\frac{f(x_n) - f(y_n)}{x_n - y_n} \right)_{n \geq 1}$$

is convergent.

118. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function such that its derivative is integrable on $[0, 1]$ and $f(1) = 0$. Prove that

$$\int_0^1 (xf'(x))^2 dx \geq 12 \left(\int_0^1 xf(x) dx \right)^2.$$

119. a) Let $a \in \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function for which there exists an antiderivative $F : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$F(x) + a \cdot f(x) \geq 0 \quad \text{for all } x \in \mathbb{R},$$

and

$$\lim_{|x| \rightarrow \infty} \frac{F(x)}{e^{|\alpha \cdot x|}} = 0 \quad \text{for all } \alpha \in \mathbb{R}^*.$$

Prove that $F(x) \geq 0$ for all $x \in \mathbb{R}$.

- b) Let $n \geq 2$ be a positive integer, $g \in \mathbb{R}[X]$, $g = X^n + a_1X^{n-1} + \cdots + a_{n-1}X + a_n$ be a polynomial with all roots real, and $f : \mathbb{R} \rightarrow \mathbb{R}$ a polynomial function such that

$$f(x) + a_1f'(x) + a_2f''(x) + \cdots + a_nf^{(n)}(x) \geq 0 \quad \text{for all } x \in \mathbb{R}.$$

Prove that $f(x) \geq 0$ for all $x \in \mathbb{R}$.

120. (2026) Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that for every $n \in \mathbb{N}^+$, there exist functions $f_n, g_n : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f_n(x) < h(x) < g_n(x)$ and $\lim_{n \rightarrow \infty} (g_n(x) - f_n(x)) = 0$ for all $x \in \mathbb{R}$.

- a) Assume that f_n and g_n are continuous for all n . Prove that h is continuous.
b) Assume that f_n and g_n are differentiable for all n . Does it follow that h is differentiable?

121. Let $f : [0, \infty) \rightarrow [0, \infty)$ be a convex and differentiable function. Suppose there exists $s > 1$ such that

$$x^s f'(x) \geq (f(f(x)))^s \quad \text{for all } x \geq 0.$$

- a) Prove that $f(x) \leq x$ for all $x \geq 0$.
b) Prove that $x^t f'(x) \geq (f(f(x)))^t$ for all $x \geq 0$ and $t \geq 1$.

122. Let a_n be the number of lattice points on the closed disk D_n of radius $n \in \mathbb{N}^+$ centered at $(0, 0)$. Determine

$$\lim_{n \rightarrow \infty} \frac{a_n}{n^2}.$$

123. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Prove that f is strictly monotone on $\mathbb{R}$ if and only if

$$\int_x^y f(t)dt \neq (y-x)f(x) \quad \text{for all } x, y \in \mathbb{R} \text{ with } x \neq y.$$